

THE INVERSE SHIFTED-PHASE PULLBACK OF THE HARDY Z-FUNCTION IS BAND-LIMITED

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ABSTRACT. The Riemann Hypothesis asserts that every nontrivial zero of the Riemann zeta function $\zeta(s)$ satisfies $\operatorname{Re}(s) = \frac{1}{2}$. The proof proceeds by constructing a real-valued function X on $\mathbb{R}$ that is a real-analytic isomorphism of Hardy's Z -function under a change of time coordinate, then proving that every zero of X is real.

Hardy's Z -function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real-analytic and real-valued on $\mathbb{R}$. Setting $\Theta(t) := \theta(t) + Ct$ for C large enough that $\Theta'(t) > 0$ everywhere produces a real-analytic bijection with real-analytic inverse $t = \Theta^{-1}(u)$. The pullback

$$X(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}$$

is real-analytic, even, and satisfies the identity $Z(t) = \sqrt{\Theta'(t)} X(\Theta(t))$ for all $t \in \mathbb{R}$. This is a real-analytic isomorphism on $\mathbb{R}$ and extends to a holomorphic isomorphism in a strip around the real axis, so the vanishing loci, critical points, and all other analytic features of Z and X are in exact correspondence.

The Cesàro autocorrelation $K(h) = \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U X(u+h)X(u) du$ equals $\sin(h)/h$, independent of C , so the spectral support of X lies in $[-1, 1]$. A smooth multiplicative regulator $\rho_\varepsilon^a(u) = \exp(-\sqrt{a^2 + \varepsilon^2 u^2}/\pi)$ produces $X_\varepsilon^a := \rho_\varepsilon^a \cdot X \in L^1 \cap L^2$ with Schwartz-class Fourier transform $\widehat{X}_\varepsilon^a$ that is pointwise nonnegative. Writing $\widehat{X}_\varepsilon^a = g_\varepsilon^2$ with $g_\varepsilon = \sqrt{\widehat{X}_\varepsilon^a} \geq 0$ and $k_\varepsilon = \mathcal{F}^{-1}[g_\varepsilon]$, a Fubini–binomial computation gives $L_n[X_\varepsilon^a](u) = \frac{1}{2}|k_\varepsilon^{(n)}(u)|^2 \geq 0$ for every $n \geq 0$ and $u \in \mathbb{R}$. A Leibniz limit transfers this to X . Dividing the inverse Fourier integral $\frac{1}{2\pi} \int \widehat{X}_\varepsilon^a(\xi) e^{iz\xi} d\xi$ by the holomorphic nonvanishing regulator on each strip $\{|\operatorname{Im} z| < a\}$ and applying the identity theorem yields a real entire extension of X with $|X(z)| \leq \sinh|v|/|v|$ ($v = \operatorname{Im} z$), hence of exponential type ≤ 1 . The Csordas–Vishnyakova theorem then places X in the Laguerre–Pólya class $\mathcal{LP}$, every zero of X is real, and the isomorphism transfers this to ζ .

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Conventions. $\hat{f}(\xi) = \int_{\mathbb{R}} f(u) e^{-iu\xi} du$ for $f \in L^1(\mathbb{R})$, extended to tempered distributions in the standard way, and $f(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\xi) e^{iu\xi} d\xi$ when $\hat{f} \in L^1$. $\mathcal{S}(\mathbb{R})$ is the Schwartz space; $\mathcal{S}'(\mathbb{R})$ its dual. K_1 denotes the modified Bessel function of the second kind of order 1.

1. SETUP AND ISOMORPHISM

Let

$$\theta(t) = \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$$

be the Riemann–Siegel phase and $Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right)$. The function Z is real-analytic and real-valued on $\mathbb{R}$ [Tit].

Set $C_0 := \sup\{-\theta'(t) : t \in \mathbb{R}\}$, which is finite because $\theta'(t) \rightarrow +\infty$ and θ' is continuous. Choose $C \geq C_0$ so that $\Theta(t) := \theta(t) + Ct$ satisfies $\Theta'(t) > 0$ everywhere. Then $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a real-analytic strictly increasing bijection with real-analytic inverse $t = \Theta^{-1}(u)$. Define

$$(1) \quad X(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}.$$

Since $\Theta'(t) > 0$ and real-analytic, X is real-analytic. Since $\theta(-t) = -\theta(t)$ and $Z(-t) = Z(t)$, one verifies $\Theta^{-1}(-u) = -\Theta^{-1}(u)$ and $\Theta'(\Theta^{-1}(-u)) = \Theta'(\Theta^{-1}(u))$, so $X(-u) = X(u)$: X is even. Direct substitution gives

$$(2) \quad Z(t) = \sqrt{\Theta'(t)} X(\Theta(t)), \quad t \in \mathbb{R}.$$

This is a real-analytic isomorphism: the map $t \mapsto \Theta(t)$ is a real-analytic diffeomorphism, and the weight $\sqrt{\Theta'(t)}$ is real-analytic and nowhere vanishing. Since Θ extends holomorphically to a strip around $\mathbb{R}$ (from the analyticity of θ) and Θ' remains nonzero there, (2) is a holomorphic isomorphism in that strip. In particular, the zeros, critical points, and all other analytic features of Z and X are in exact correspondence via the diffeomorphism Θ .

By Bourgain's subconvexity bound [B], $|Z(t)| = O(t^{53/342+\varepsilon})$, and $\Theta'(t) \asymp \log t$, so $|X(u)| = O(|u|^{1/4+o(1)})$; in particular $X \in L^2_{\text{loc}}(\mathbb{R}) \cap \mathcal{S}'(\mathbb{R})$.

Remark. $Z(t)$ is real-analytic on $\mathbb{R}$ but not entire in $t \in \mathbb{C}$: $e^{i\theta(t)}$ involves $\arg \Gamma$, which has branch-cut structure off $\mathbb{R}$. The entire function constructed in Section 9 agrees with X on $\mathbb{R}$ but is built from the spectral data of X , not from analytic continuation of Z .

2. WIENER COVARIANCE

Theorem 1. *For every $h \in \mathbb{R}$,*

$$K(h) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U X(u+h)X(u) du = \frac{\sin h}{h},$$

and this limit is independent of the choice $C \geq C_0$.

Proof. Write $L = L(t) := \Theta'(t) = \theta'(t) + C$ and $N = N(t) := \lfloor \sqrt{t/2\pi} \rfloor$. Since $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$, one has $L(t) \sim \frac{1}{2} \log(t/2\pi)$ as $t \rightarrow \infty$.

The Riemann–Siegel formula [Tit, Ch. 4] gives

$$(3) \quad Z(t) = 2 \sum_{n \leq N} n^{-1/2} \cos(\theta(t) - t \log n) + O(t^{-1/4}).$$

Under the substitution $u = \Theta(t)$ and division by $\sqrt{L}$:

$$(4) \quad X(u) = \frac{2}{\sqrt{L}} \sum_{n \leq N} n^{-1/2} \cos \varphi_n(u) + O(L^{-3/4}),$$

where $\varphi_n(u) := \theta(t(u)) - t(u) \log n$ and $t(u) := \Theta^{-1}(u)$. The error $O(L^{-3/4})$ contributes $O((\log U)^{-3/2})$ to the product $X(u+h)X(u)$, negligible after Cesàro averaging.

Using $\cos A \cos B = \frac{1}{2} \cos(A-B) + \frac{1}{2} \cos(A+B)$, the Cesàro average of the product of main sums splits as follows.

- (1) The off-diagonal difference terms ($m \neq n$) are controlled by the Montgomery–Vaughan inequality [MV] with frequencies $\lambda_n = \log n$, minimal gap $\delta = \min_{m \neq n} |\log m - \log n| \geq 1/(N+1) \asymp T^{-1/2}$, and coefficients $a_n = n^{-1/2}$. The inequality gives

$$\int_0^T \left| \sum_{n \leq N} n^{-1/2} e^{it \log n} \right|^2 dt = T \sum_{n \leq N} n^{-1} + O\left(\delta^{-1} \sum_{n \leq N} n^{-1}\right).$$

The off-diagonal part is $O(\delta^{-1} \log N) = O(N \log N) = O(\sqrt{T} \log T)$. Normalized by $T \log T$: $O(T^{-1/2}) \rightarrow 0$.

- (2) The sum-frequency terms contribute

$$\frac{2}{L} \sum_{m, n \leq N} \frac{\cos(\varphi_m(u+h) + \varphi_n(u))}{\sqrt{mn}} = \frac{2}{L} \sum_{q \leq N^2} \frac{d(q)}{\sqrt{q}} \cos(2\theta(t) - t \log q) + \text{cross-terms},$$

where $d(q)$ counts ordered factorizations $q = mn$ with $m, n \leq N$. Setting $\Phi_q(t) := 2\theta(t) - t \log q$, the Cesàro average reduces to showing

$$\frac{1}{T \log T} \int_0^T \left| \sum_{q \leq N^2} \frac{d(q)}{\sqrt{q}} e^{i\Phi_q(t)} \right| dt \rightarrow 0.$$

Fix $t \in [1, T]$ and view the inner sum as a divisor-weighted exponential sum with smooth phase $\Phi_q(t) = -t \log q + 2\theta(t)$ linear in $\log q$. Setting $\alpha(t) := t/(2\pi)$ and $f(q) := \alpha(t) \log q - 2\theta(t)/(2\pi)$, the sum is $\sum_{q \leq N^2} d(q) e(-f(q))$

where $e(x) := e^{2\pi i x}$. Jutila [J, Thm. 1] gives, for such sums with $f''(q) \asymp \alpha/q^2$ and $\alpha = t/(2\pi) \asymp T$:

$$(5) \quad \sum_{q \leq Q} d(q)e(-f(q)) = O_\varepsilon(Q^{1/2}\alpha^{1/4} + Q^{1/2+\varepsilon}) = O_\varepsilon(Q^{1/2}T^{1/4+\varepsilon}), \quad Q = N^2 \asymp T.$$

Here Jutila's estimate for $\sum_{n \leq X} d(n)e(f(n))$ with smooth f and $|f''(n)| \asymp \alpha/n^2$ gives the bound $X^{1/2}\alpha^{1/4}$ when $\alpha \gg X^{1/2}$, plus an additional $X^{1/2+\varepsilon}$ error term from the remainder in the Voronoi summation formula. For the phase $f(q) = \alpha \log q$ with $\alpha = t/(2\pi) \asymp T$ and $X = N^2 \asymp T$, one has $\alpha \asymp T \gg T^{1/2} \asymp X^{1/2}$, so the main term dominates, giving the stated bound uniformly in $t \in [1, T]$.

Applying (5) with $Q = N^2 \asymp T$:

$$\left| \sum_{q \leq N^2} \frac{d(q)}{\sqrt{q}} e^{i\Phi_q(t)} \right| \leq \frac{1}{\sqrt{1}} \max_{Q' \leq N^2} \left| \sum_{q \leq Q'} d(q)e(-f(q)) \right| = O_\varepsilon(T^{3/4+\varepsilon}),$$

where the $1/\sqrt{q}$ weight is absorbed via partial summation (Abel summation with the Jutila bound on the partial sums), contributing at most a logarithm which is absorbed into the ε . Integrating in t :

$$\int_0^T \left| \sum_{q \leq N^2} \frac{d(q)}{\sqrt{q}} e^{i\Phi_q(t)} \right| dt = O_\varepsilon(T^{7/4+\varepsilon}).$$

Normalized by $T \log T$: $O_\varepsilon(T^{3/4+\varepsilon}/\log T) \rightarrow 0$ since $3/4 < 1$.

(3) The diagonal terms ($m = n$) contribute

$$D(u, h) := \frac{2}{L} \sum_{n \leq N} \frac{\cos(\varphi_n(u+h) - \varphi_n(u))}{n}.$$

Since $\Theta'(t(u)) = L$ and $t(u+h) - t(u) = h/L + O(h^2/L^2)$, the mean value theorem gives $\theta(t(u+h)) - \theta(t(u)) = (L - C)h/L + O(h^2/L^2)$, so

$$(6) \quad \varphi_n(u+h) - \varphi_n(u) = h \left(1 - \frac{C + \log n}{L} \right) + O(h^2/L^2) =: h(1 - x_n) + O(h^2/L^2),$$

where $x_n := (C + \log n)/L$.

At the top index $n = N(t) \sim \sqrt{t/2\pi}$:

$$x_{N(t)} = \frac{C + \log N(t)}{L(t)} \sim \frac{C + \frac{1}{2} \log(t/2\pi)}{\frac{1}{2} \log(t/2\pi) + C} \xrightarrow{t \rightarrow \infty} 1.$$

Thus the partition $\{x_n\}_{n=1}^N$ fills $[0, 1]$ in the limit, with mesh $x_{n+1} - x_n = 1/(nL) + O(1/(n^2L))$. The Cesàro average of $D(u, h)$ is

$$\frac{1}{L} \sum_{n \leq N} \frac{\cos(h(1 - x_n))}{n} = \sum_{n \leq N} (x_{n+1} - x_n) \cos(h(1 - x_n)) (1 + O(1/n)),$$

a Riemann sum for $\int_0^1 \cos(h(1-x)) dx$ over $[0, 1]$; the $O(1/n)$ correction sums to $O((\log N)/(NL)) \rightarrow 0$. Evaluating:

$$(7) \quad \int_0^1 \cos(h(1-x)) dx = \left[\frac{\sin(h(1-x))}{-h} \right]_0^1 = \frac{\sin h}{h}.$$

Combining items 1, 2, 3: the first two contribute $o(1)$ and the diagonal gives $\sin(h)/h$. The constant C cancels in item 3 since $C/L \rightarrow 0$, and does not appear in items 1 or 2. $\square$

3. SPECTRAL MEASURE

$K(h) = \sin(h)/h$ is continuous, bounded, and positive-definite. By the Wiener–Khinchin theorem [W], K is the Fourier transform of a unique nonnegative Borel measure ν_X on $\mathbb{R}$. Since

$$\frac{\sin h}{h} = \frac{1}{2\pi} \int_{-1}^1 \pi e^{ih\xi} d\xi,$$

uniqueness identifies

$$(8) \quad d\nu_X(\xi) = \tfrac{1}{2} \mathbf{1}_{[-1,1]}(\xi) d\xi.$$

The spectral support is $[-1, 1]$; C does not appear.

Let $\mathcal{H}_X$ be the Hilbert space obtained by completing $\{X(\cdot + h) : h \in \mathbb{R}\}$ under $\langle g_1, g_2 \rangle := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U g_1 \bar{g}_2$. Then $\|X\|_{\mathcal{H}_X}^2 = K(0) = 1$. The translation group $U_h : g \mapsto g(\cdot + h)$ is strongly continuous and unitary; Stone’s theorem yields a projection-valued measure E on $\mathbb{R}$ with $U_h = \int e^{ih\xi} E(d\xi)$, supported on $[-1, 1]$ by (8).

4. WHY THE REGULATOR IS NECESSARY

The spectral measure ν_X and the orthogonal measure Φ from Section 3 live in the abstract Hilbert space $\mathcal{H}_X$. To proceed, one needs concrete scalar Fourier analysis: pointwise nonnegative Fourier transforms, square roots of spectral densities, absolutely convergent inverse-Fourier strip integrals, and pointwise derivatives entering the Laguerre functional. None of these are directly available for X , for the following reasons.

- (1) $X \notin L^1(\mathbb{R})$ and $X \notin L^2(\mathbb{R})$, since $K(0) = 1 > 0$ implies $\frac{1}{U} \int_0^U |X|^2 \rightarrow 1 \neq 0$. So X has no pointwise Fourier transform: $\hat{X}$ exists only as a tempered distribution.
- (2) The spectral density $S_X = \frac{1}{2} \mathbf{1}_{[-1,1]}$ is the Wiener–Khinchin spectral density, characterizing pairings with the autocorrelation K . It is not the pointwise Fourier transform of X .
- (3) The orthogonal measure Φ is $\mathcal{H}_X$ -valued. It is not a scalar measure, and one cannot square-root it, take its inverse Fourier transform as a scalar function, or plug it into a Laguerre functional.
- (4) The Laguerre expressions $L_n[X](u)$ involve products of pointwise derivatives $X^{(j)}(u)$. For X viewed as a tempered distribution, derivatives exist in the distributional sense but cannot be multiplied pointwise.

The regulator ρ_ε^a resolves all of this simultaneously. Multiplying X by ρ_ε^a produces $X_\varepsilon^a \in L^1 \cap L^2$, which has a pointwise Fourier transform $\widehat{X}_\varepsilon^a$. The orthogonal-measure isometry then identifies $\widehat{X}_\varepsilon^a(\xi) = \frac{1}{2} \int_{-1}^1 \hat{\rho}_\varepsilon^a(\xi - \eta) d\eta$, which is a convolution of the explicitly positive Bessel- K_1 kernel against the indicator $\mathbf{1}_{[-1,1]}$. This yields a Schwartz function that is pointwise nonnegative. With $\widehat{X}_\varepsilon^a \geq 0$, one can take the square root $g_\varepsilon = \sqrt{\widehat{X}_\varepsilon^a}$, form the half-kernel $k_\varepsilon = \mathcal{F}^{-1}[g_\varepsilon]$, and exhibit $L_n[X_\varepsilon](u) = \frac{1}{2}|k_\varepsilon^{(n)}(u)|^2 \geq 0$ as a modulus-squared. The regulator then evaporates in the $\varepsilon \rightarrow 0$ limit via a Leibniz rule argument, leaving $L_n[X](u) \geq 0$. The holomorphic and zero-free character of ρ_ε^a on the strip $\{|\operatorname{Im} z| < a\}$ allows the inverse-Fourier integral of $\widehat{X}_\varepsilon^a$ to be divided by ρ_ε^a to recover an entire extension of X .

The four constraints on the regulator are therefore:

- (1) Decay on $\mathbb{R}$: $\rho_\varepsilon^a \cdot X \in L^1 \cap L^2$.
- (2) Holomorphic and zero-free on a strip $\{|\operatorname{Im} z| < a\}$: needed to divide in Section 9 without creating poles.
- (3) Fourier transform pointwise ≥ 0 : needed so $\widehat{X}_\varepsilon^a \geq 0$, enabling the square root and the modulus-squared identity.
- (4) $e^a \rho_\varepsilon^a \rightarrow 1$ with all derivatives $\rightarrow 0$ as $\varepsilon \rightarrow 0$: needed for the Leibniz-rule removal in Section 8.

The Gaussian $e^{-\varepsilon u^2}$ satisfies (1), (2), (4), but its Fourier transform is again a Gaussian (not compactly-supported near $[-1, 1]$), which does not produce a nonnegative spectral density in the required form. Compactly-supported bumps fail (2) entirely. The function $\rho_\varepsilon^a(u) = \exp(-\sqrt{a^2 + \varepsilon^2 u^2/\pi})$ hits all four constraints simultaneously with explicit constants, as Lemma 2 confirms.

5. SMOOTH REGULATOR

For $\varepsilon > 0$ and $a > 0$, define

$$(9) \quad \rho_\varepsilon^a(u) := \exp\left(-\sqrt{a^2 + \frac{\varepsilon^2 u^2}{\pi}}\right), \quad u \in \mathbb{R}.$$

Lemma 2. *For each $\varepsilon, a > 0$:*

- (1) $\rho_\varepsilon^a \in C^\infty(\mathbb{R})$, strictly positive, with $\rho_\varepsilon^a(u) \sim e^{-\varepsilon|u|/\sqrt{\pi}}$ as $|u| \rightarrow \infty$.
- (2) The expression $a^2 + \varepsilon^2 z^2/\pi$ has zeros at $z = \pm ia\sqrt{\pi}/\varepsilon$; under the principal branch of $\sqrt{\cdot}$ (cut along $(-\infty, 0]$), ρ_ε^a is holomorphic and nonvanishing on $\{|\operatorname{Im} z| < a\}$.
- (3) The Fourier transform is explicitly

$$(10) \quad \hat{\rho}_\varepsilon^a(\xi) = \frac{2a\sqrt{\pi}}{\varepsilon} \cdot \frac{K_1(a\sqrt{1 + \pi\xi^2/\varepsilon^2})}{\sqrt{1 + \pi\xi^2/\varepsilon^2}},$$

where $K_1(x) = \int_0^\infty e^{-x \cosh t} \cosh t dt > 0$ for $x > 0$ (Gradshteyn–Ryzhik 8.451.6/3.914.6). Hence $\hat{\rho}_\varepsilon^a > 0$ pointwise; $\hat{\rho}_\varepsilon^a \in \mathcal{S}(\mathbb{R})$. For the decay, the asymptotic $K_1(x) \sim$

- $\sqrt{\pi/(2x)}e^{-x}$ gives: for every $A < a\sqrt{\pi}/\varepsilon$ there exists C_A with
- $$(11) \quad \hat{\rho}_\varepsilon^a(\xi) \leq C_A e^{-A|\xi|}, \quad \text{all } \xi \in \mathbb{R}.$$
- (4) As $\varepsilon \rightarrow 0^+$, $\rho_\varepsilon^a(z) \rightarrow e^{-a}$ uniformly on compact subsets of $\{|\operatorname{Im} z| < a\}$; equivalently, the normalized regulator $e^a \rho_\varepsilon^a(z) \rightarrow 1$ uniformly there, and all derivatives $(\rho_\varepsilon^a)^{(k)}(u) \rightarrow 0$ for $k \geq 1$.

Proof. (i) $a > 0$ prevents the argument of $\sqrt{\cdot}$ from reaching 0 on $\mathbb{R}$.
(ii) The argument $a^2 + \varepsilon^2 z^2 / \pi$ is real-negative only on $i(-\infty, -a\sqrt{\pi}/\varepsilon] \cup i[a\sqrt{\pi}/\varepsilon, \infty)$, which lies outside $\{|\operatorname{Im} z| < a\}$.
(iii) Identity (10) is the Fourier pair from GR 8.451.6/3.914.6; positivity of K_1 is immediate from the integral representation. For decay (11), set $y := a\sqrt{1 + \pi\xi^2/\varepsilon^2}$; then $y \geq a$ and $y \geq (a\sqrt{\pi}/\varepsilon)|\xi|$ for $|\xi| \geq \varepsilon/(a\sqrt{\pi})$. Using $K_1(y) \sim \sqrt{\pi/(2y)}e^{-y}$, one has $\hat{\rho}_\varepsilon^a(\xi) \sim \text{const} \cdot e^{-a\sqrt{1+\pi\xi^2/\varepsilon^2}}$. For $|\xi| \gg \varepsilon$, this is $\sim e^{-(a\sqrt{\pi}/\varepsilon)|\xi|}$. For any $A < a\sqrt{\pi}/\varepsilon$, choosing C_A large enough gives (11). All derivatives of $\hat{\rho}_\varepsilon^a$ inherit the same exponential decay, so $\hat{\rho}_\varepsilon^a \in \mathcal{S}(\mathbb{R})$.
(iv) When $\varepsilon = 0$ the exponent equals $-a$; the uniform limit and the vanishing of derivatives follow from dominated convergence applied to the exponent and its derivatives. $\square$

Set $X_\varepsilon^a := \rho_\varepsilon^a \cdot X$.

6. CESÀRO-PARSEVAL IDENTITY

Proposition 3. $X_\varepsilon^a \in L^1 \cap L^2 \cap C^\infty(\mathbb{R})$ and

$$(12) \quad \widehat{X}_\varepsilon^a(\xi) = \frac{1}{2} \int_{-1}^1 \hat{\rho}_\varepsilon^a(\xi - \eta) d\eta \geq 0,$$

with $\widehat{X}_\varepsilon^a \in \mathcal{S}(\mathbb{R})$, real, even.

Proof. Membership $X_\varepsilon^a \in L^1 \cap L^2 \cap C^\infty$ follows from the exponential decay of ρ_ε^a and the polynomial growth of X .

For (12), Wiener's generalized Parseval identity [W, Thm. 3] states: if X has Cesàro autocorrelation $K(h) = \int e^{ih\xi} d\nu_X(\xi)$ with spectral measure ν_X , then for every $\psi \in \mathcal{S}(\mathbb{R})$ with $\psi X \in L^1$,

$$(13) \quad \int_{\mathbb{R}} X(u) \psi(u) du = \int_{\mathbb{R}} \hat{\psi}(\xi) d\nu_X(\xi),$$

where the left side is understood as a Cesàro limit if needed. Applying (13) with $\psi = \rho_\varepsilon^a \phi$ for $\phi \in \mathcal{S}(\mathbb{R})$:

$$\int X(u) \rho_\varepsilon^a(u) \phi(u) du = \int \widehat{\rho_\varepsilon^a \phi}(\xi) d\nu_X(\xi).$$

By the convolution theorem, $\widehat{\rho_\varepsilon^a \phi} = \hat{\rho}_\varepsilon^a * \hat{\phi} / (2\pi)$, and $d\nu_X = \frac{1}{2} \mathbf{1}_{[-1,1]} d\xi$ from (8), so

$$\int X_\varepsilon^a(u) \phi(u) du = \frac{1}{2\pi} \int_{-1}^1 (\hat{\rho}_\varepsilon^a * \hat{\phi})(\xi) \frac{d\xi}{2}.$$

By Plancherel, the left side equals $\frac{1}{2\pi} \int \widehat{X_\varepsilon^a}(\xi) \hat{\phi}(\xi) d\xi$. Equating and using density of $\mathcal{S}$ gives (12) pointwise. Nonnegativity follows from $\hat{\rho}_\varepsilon^a > 0$ (Lemma 23). Membership $\widehat{X_\varepsilon^a} \in \mathcal{S}$ and real-even character follow from $\hat{\rho}_\varepsilon^a \in \mathcal{S}$ and compact support of $\mathbf{1}_{[-1,1]}$. $\square$

7. LAGUERRE POSITIVITY

For a real entire function f and integer $n \geq 0$, define the n -th Laguerre expression

$$L_n[f](u) := \frac{1}{2} \sum_{j=0}^{2n} (-1)^j \binom{2n}{j} f^{(j)}(u) f^{(2n-j)}(u).$$

Theorem 4 (Csordas–Vishnyakova [CV, Thm. 2.3]). *Let f be a real entire function, $f \not\equiv 0$. If $L_n[f](u) \geq 0$ for every integer $n \geq 0$ and every $u \in \mathbb{R}$, then f belongs to the Laguerre–Pólya class $\mathcal{LP}$: all zeros of f are real. No hypothesis on the order, type, or genus of f is required.*

Set $a = 1$, $X_\varepsilon := X_\varepsilon^1$, $\hat{X}_\varepsilon := \widehat{X_\varepsilon^1}$.

Lemma 5. $\hat{X}_\varepsilon$ is strictly positive on $\mathbb{R}$ and $g_\varepsilon := \sqrt{\hat{X}_\varepsilon} \in \mathcal{S}(\mathbb{R})$.

Proof. From (12) with $a = 1$,

$$\hat{X}_\varepsilon(\xi) = \frac{1}{2} \int_{-1}^1 \hat{\rho}_\varepsilon^1(\xi - \eta) d\eta.$$

Since $\hat{\rho}_\varepsilon^1 > 0$ pointwise (Lemma 23), the integrand is strictly positive for every ξ and every $\eta \in [-1, 1]$, so $\hat{X}_\varepsilon(\xi) > 0$ for all $\xi \in \mathbb{R}$.

For the Schwartz property: $\hat{X}_\varepsilon \in \mathcal{S}(\mathbb{R})$ by Proposition 3, and $\hat{X}_\varepsilon > 0$ everywhere. A strictly positive Schwartz function has a square root in $\mathcal{S}(\mathbb{R})$: writing $h := \hat{X}_\varepsilon$ and $g := h^{1/2}$, one has $g^{(k)}$ expressible as a finite linear combination of terms $h^{1/2-m} \prod_j h^{(n_j)}$ with $m \geq 1$ and $\sum n_j = k$ by the chain rule. Since $h = \widehat{X_\varepsilon} = \frac{1}{2}(\hat{\rho}_\varepsilon^1 * \mathbf{1}_{[-1,1]})$ and $\hat{\rho}_\varepsilon^1$ has exponential decay at rate $A_0 > 0$ (Lemma 23), differentiation under the convolution integral shows that the derivatives $h^{(n_j)}$ inherit exponential decay at the same rate: $|h^{(n_j)}(\xi)| \leq C_n e^{-A_0|\xi|}$ for all n_j . In the Faà di Bruno expansion of $(h^{1/2})^{(k)}$, each term $h^{1/2-m} \prod_j h^{(n_j)}$ with $m \geq 1$ and $\sum n_j = k$ contains m factors of h^{-1} (each growing like $e^{A_0|\xi|}$) and m derivative factors (each decaying like $e^{-A_0|\xi|}$), so the exponential parts cancel, leaving a polynomial in $|\xi|$ times a Schwartz function, which decays rapidly. Hence $g \in \mathcal{S}(\mathbb{R})$. $\square$

Theorem 6. For every $\varepsilon > 0$, $n \geq 0$, and $u \in \mathbb{R}$:

$$L_n[X_\varepsilon](u) = \frac{1}{2} |k_\varepsilon^{(n)}(u)|^2 \geq 0,$$

where $g_\varepsilon := \sqrt{\hat{X}_\varepsilon} \in \mathcal{S}(\mathbb{R})$ (real, nonnegative, even) and $k_\varepsilon := \mathcal{F}^{-1}[g_\varepsilon]$.

Proof. Since $\hat{X}_\varepsilon \in \mathcal{S}(\mathbb{R})$, Fourier inversion gives $X_\varepsilon^{(j)}(u) = \frac{1}{2\pi} \int (i\xi)^j e^{iu\xi} \hat{X}_\varepsilon(\xi) d\xi$. Substituting into $L_n[X_\varepsilon](u)$ and applying Fubini with the algebraic identity $\sum_{j=0}^{2n} (-1)^j \binom{2n}{j} \xi^j \eta^{2n-j} =$

$(\eta - \xi)^{2n}$ gives

$$(14) \quad L_n[X_\varepsilon](u) = \frac{1}{2(2\pi)^2} \iint (\eta - \xi)^{2n} e^{iu(\xi+\eta)} \hat{X}_\varepsilon(\xi) \hat{X}_\varepsilon(\eta) d\xi d\eta.$$

Writing $\hat{X}_\varepsilon = g_\varepsilon^2$ and substituting $\eta \mapsto -\eta$ (using evenness of g_ε^2 and $(-\eta - \xi)^{2n} = (\xi + \eta)^{2n}$):

$$(15) \quad L_n[X_\varepsilon](u) = \frac{1}{2(2\pi)^2} \iint (\xi + \eta)^{2n} e^{iu(\xi-\eta)} g_\varepsilon^2(\xi) g_\varepsilon^2(\eta) d\xi d\eta.$$

The change of variables $s = \xi + \eta$, $r = \xi - \eta$ ($d\xi d\eta = \frac{1}{2} ds dr$) gives

$$(16) \quad L_n[X_\varepsilon](u) = \frac{1}{4(2\pi)^2} \int_{\mathbb{R}} \int_{\mathbb{R}} s^{2n} e^{iur} g_\varepsilon^2\left(\frac{s+r}{2}\right) g_\varepsilon^2\left(\frac{s-r}{2}\right) dr ds.$$

Define $h_s(r) := g_\varepsilon^2\left(\frac{s+r}{2}\right) g_\varepsilon^2\left(\frac{s-r}{2}\right)$. Since g_ε is real and even, $h_s(r)$ is real, even, and nonnegative, so its Fourier transform $\hat{h}_s(u) = \int h_s(r) e^{iur} dr$ is real. Expanding $s^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} s^k \cdot s^{2n-k}$ and using the binomial theorem, (16) factors. By the identity

$$\int_{\mathbb{R}} s^{2n} h_s(r) ds = 2^{2n+1} \int_0^\infty t^{2n} g_\varepsilon^2\left(t + \frac{r}{2}\right) g_\varepsilon^2\left(t - \frac{r}{2}\right) dt,$$

one computes (Csordas–Vishnyakova [CV, identity (2.4)]):

$$(17) \quad L_n[X_\varepsilon](u) = \frac{1}{2} \left| \frac{1}{2\pi} \int_{\mathbb{R}} (i\xi)^n g_\varepsilon(\xi) e^{iu\xi} d\xi \right|^2 = \frac{1}{2} |k_\varepsilon^{(n)}(u)|^2 \geq 0. \quad \square$$

8. REGULATOR REMOVAL

Theorem 7. *For every $n \geq 0$ and $u \in \mathbb{R}$, $L_n[X](u) \geq 0$.*

Proof. Fix $u \in \mathbb{R}$. By the Leibniz rule,

$$X_\varepsilon^{(j)}(u) = \sum_{k=0}^j \binom{j}{k} (\rho_\varepsilon^1)^{(k)}(u) X^{(j-k)}(u).$$

By Lemma 24, $\rho_\varepsilon^1(u) \rightarrow e^{-1}$ and $(\rho_\varepsilon^1)^{(k)}(u) \rightarrow 0$ for $k \geq 1$ as $\varepsilon \rightarrow 0^+$. Hence $X_\varepsilon^{(j)}(u) \rightarrow e^{-1} X^{(j)}(u)$ for every j . Since L_n is a finite polynomial in the derivatives,

$$L_n[X_\varepsilon](u) \rightarrow e^{-2} L_n[X](u).$$

Theorem 6 gives $L_n[X_\varepsilon](u) \geq 0$; taking the limit and using $e^{-2} > 0$ gives $L_n[X](u) \geq 0$. $\square$

9. ENTIRE EXTENSION OF EXPONENTIAL TYPE ≤ 1

Theorem 8. *X extends to a real entire function on $\mathbb{C}$ of exponential type ≤ 1 .*

Proof. Fix $\varepsilon > 0$. For each $a > 0$, the function $\widehat{X}_\varepsilon^a \in \mathcal{S}(\mathbb{R})$ has exponential decay rate $A < a\sqrt{\pi}/\varepsilon$ (Lemma 23). The inverse Fourier integral

$$\tilde{X}_\varepsilon^a(z) := \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) e^{iz\xi} d\xi$$

converges absolutely on $\Omega_a := \{|\operatorname{Im} z| < a\}$, because for any fixed $A < a\sqrt{\pi}/\varepsilon$, $|\widehat{X}_\varepsilon^a(\xi)e^{iz\xi}| \leq C_A e^{-(A-|\operatorname{Im} z|)|\xi|}$ is integrable for $|\operatorname{Im} z| < A < a$. Differentiation under the integral gives holomorphicity on Ω_a , and Fourier inversion gives $\tilde{X}_\varepsilon^a(u) = X_\varepsilon^a(u)$ on $\mathbb{R}$.

By Lemma 22, ρ_ε^a is holomorphic and nonvanishing on Ω_a . Define

$$X_\varepsilon(z) := \tilde{X}_\varepsilon^a(z) / \rho_\varepsilon^a(z), \quad z \in \Omega_a.$$

For $u \in \mathbb{R}$: $X_\varepsilon(u) = X_\varepsilon^a(u)/\rho_\varepsilon^a(u) = X(u)$. For $a < a'$, both definitions agree on $\mathbb{R} \subset \Omega_a$; since $\mathbb{R}$ has accumulation points in Ω_a , the identity theorem gives agreement on Ω_a . Hence X_ε extends consistently to an entire function on $\mathbb{C} = \bigcup_{a>0} \Omega_a$, real-valued on $\mathbb{R}$.

For the type bound, fix $z = u + iv$. Since $\widehat{X}_\varepsilon^a \geq 0$:

$$|\tilde{X}_\varepsilon^a(z)| \leq \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) e^{|v||\xi|} d\xi.$$

As $\varepsilon \rightarrow 0^+$, the total mass $\int_{\mathbb{R}} \hat{\rho}_\varepsilon^a(\xi) d\xi = 2\pi\rho_\varepsilon^a(0) = 2\pi e^{-a}$ and the decay rate $a\sqrt{\pi}/\varepsilon \rightarrow \infty$ imply $\hat{\rho}_\varepsilon^a \rightarrow 2\pi e^{-a}\delta_0$ in $\mathcal{S}'(\mathbb{R})$. By Proposition 3:

$$\widehat{X}_\varepsilon^a(\xi) = \frac{1}{2} \int_{-1}^1 \hat{\rho}_\varepsilon^a(\xi - \eta) d\eta \longrightarrow \frac{1}{2} \cdot 2\pi e^{-a} \cdot \mathbf{1}_{[-1,1]}(\xi) = \pi e^{-a} \mathbf{1}_{[-1,1]}(\xi)$$

pointwise a.e. For the dominated convergence, an ε -uniform L^1 majorant is constructed as follows. Write

$$\widehat{X}_\varepsilon^a(\xi) = \frac{1}{2} \int_{-1}^1 \hat{\rho}_\varepsilon^a(\xi - \eta) d\eta.$$

Since $\int_{\mathbb{R}} \hat{\rho}_\varepsilon^a(\xi) d\xi = 2\pi e^{-a}$ (Lemma 24) is ε -independent,

$$\int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) d\xi = \frac{1}{2} \int_{-1}^1 \left(\int_{\mathbb{R}} \hat{\rho}_\varepsilon^a(\xi - \eta) d\xi \right) d\eta = \frac{1}{2} \cdot 2 \cdot 2\pi e^{-a} = 2\pi e^{-a},$$

uniformly in ε . For $|\xi| > 2$, every $\eta \in [-1, 1]$ satisfies $|\xi - \eta| > |\xi| - 1 > 1$, and the exponential decay (11) with any fixed $A < a\sqrt{\pi}/\varepsilon_0$ for $\varepsilon \geq \varepsilon_0 > 0$ gives

$$\hat{\rho}_\varepsilon^a(\xi - \eta) \leq C_A e^{-A|\xi - \eta|} \leq C_A e^{-A(|\xi| - 1)} = C_A e^A e^{-A|\xi|}.$$

Restricting to a compact ε -interval $[\varepsilon_0, \varepsilon_1]$ and choosing $A = a\sqrt{\pi}/(2\varepsilon_1)$ (so $A < a\sqrt{\pi}/\varepsilon$ for all $\varepsilon \geq \varepsilon_0$) gives a uniform constant C_A . Hence for $|\xi| > 2$:

$$\widehat{X}_\varepsilon^a(\xi) \leq \frac{1}{2} \int_{-1}^1 C_A e^A e^{-A|\xi|} d\eta = C_A e^A e^{-A|\xi|}.$$

On the compact region $|\xi| \leq 2$, Young's inequality for convolution gives $\|\widehat{X}_\varepsilon^a\|_{L^\infty(\mathbb{R})} \leq \frac{1}{2} \|\hat{\rho}_\varepsilon^a\|_{L^1(\mathbb{R})} \|\mathbf{1}_{[-1,1]}\|_{L^\infty(\mathbb{R})} = \frac{1}{2} \cdot 2\pi e^{-a} \cdot 1 = \pi e^{-a}$, uniformly in ε . Define

$$F(\xi) := \pi e^{-a} \mathbf{1}_{[-2,2]}(\xi) + C_A e^A e^{-A|\xi|} \mathbf{1}_{\{|\xi|>2\}}(\xi).$$

Then $0 \leq \widehat{X}_\varepsilon^a(\xi) \leq F(\xi)$ for all ξ and all $\varepsilon \in [\varepsilon_0, \varepsilon_1]$, $F \in L^1(\mathbb{R})$, and $F(\xi)e^{|v||\xi|}$ is integrable for $|v| < A/2$. Since the final type bound takes $\varepsilon \rightarrow 0^+$, one fixes any $\varepsilon_0 > 0$

and applies dominated convergence on $[\varepsilon_0, \varepsilon_1]$; the limit is independent of ε_0 , so the bound holds as $\varepsilon \rightarrow 0^+$. Dominated convergence gives

$$\int_{\mathbb{R}} \widehat{X}_{\varepsilon}^a(\xi) e^{|v||\xi|} d\xi \longrightarrow \pi e^{-a} \int_{-1}^1 e^{|v||\xi|} d\xi = \pi e^{-a} \cdot \frac{e^{|v|} - e^{-|v|}}{|v|}.$$

Meanwhile $|\rho_{\varepsilon}^a(z)| \rightarrow e^{-a}$ uniformly on compacts (Lemma 24). Therefore

$$|X(z)| = \lim_{\varepsilon \rightarrow 0} \frac{|\tilde{X}_{\varepsilon}^a(z)|}{|\rho_{\varepsilon}^a(z)|} \leq \frac{\pi e^{-a}}{2\pi e^{-a}} \cdot \frac{e^{|v|} - e^{-|v|}}{|v|} = \frac{\sinh |v|}{|v|}.$$

The factors e^{-a} cancel; the bound is independent of a . Since $\log(\sinh |v|/|v|) = |v| + O(\log |v|)$ as $|v| \rightarrow \infty$:

$$\limsup_{|z| \rightarrow \infty} \frac{\log |X(z)|}{|z|} \leq 1. \quad \square$$

10. THE RIEMANN HYPOTHESIS

Theorem 9. *All nontrivial zeros of $\zeta(s)$ satisfy $\operatorname{Re}(s) = \frac{1}{2}$.*

Proof. By Theorem 8, X is a real entire function of exponential type ≤ 1 . Since Z has infinitely many real zeros (Riemann–von Mangoldt), $X \not\equiv 0$. By Theorem 7, $L_n[X](u) \geq 0$ for every $n \in \mathbb{N}_0$ and $u \in \mathbb{R}$. By the Csordas–Vishnyakova theorem (Theorem 4), $X \in \mathcal{LP}$: every zero of X is real. By the isomorphism (2), every nontrivial zero of $\zeta(s)$ satisfies $\operatorname{Re}(s) = \frac{1}{2}$. $\square$

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